

Speech Synthesis

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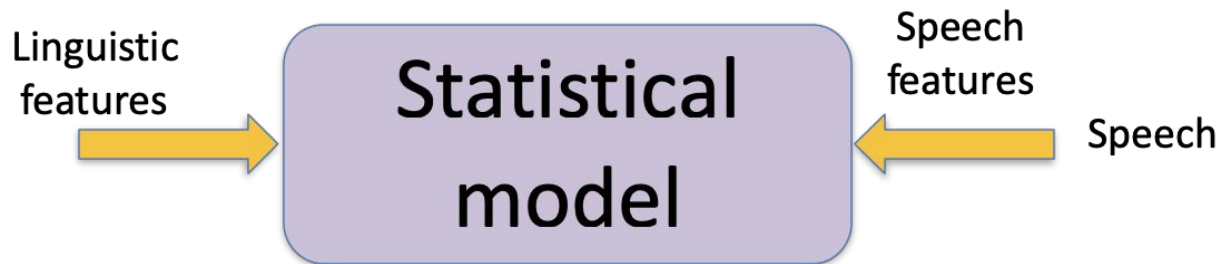
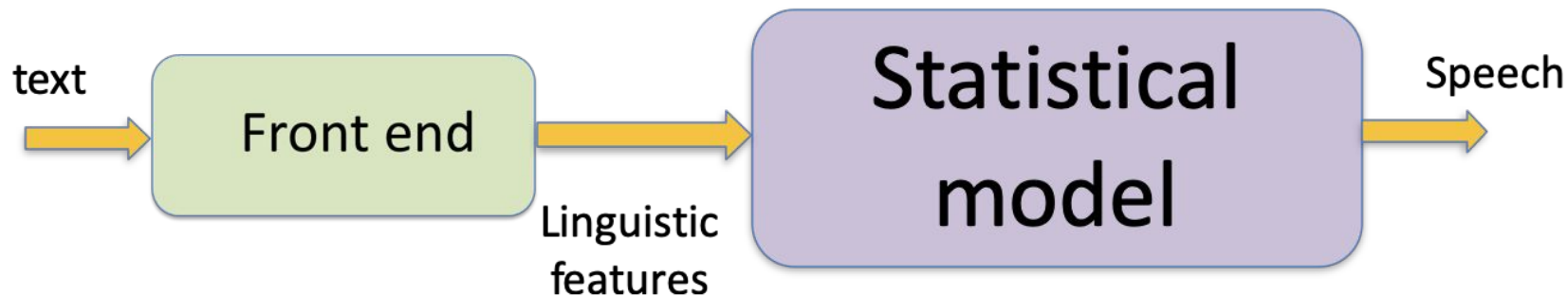
Outline

- Basic introduction to Speech Synthesis
- Components of TTS
- Traditional Methods
- Advanced TTS Systems
- Developing TTS Systems
- Future Directions

Methods

- Formant Speech Synthesis (1960s-1980s): Klatt synthesizer
- Concatenative Speech Synthesis (1990s - Early 2000s): Festival
- Statistical Parametric Speech Synthesis
 - HMM based Speech Synthesis (Early 2000s - Mid 2010s): HTS
 - DNN Based Speech Synthesis (2013s -2017s): Merlin
- Advanced Speech synthesis
 - Deep neural networks (Transformer, RNN, CNNs) (2017 - present): Tacotron 2, WaveNet
 - Architectures like flow/Diffusion based models: (2020-present) Grad-TTS (2021), DiffSinger (2022), StyleTTS2 (2023)
 - LLM Based models (2022 - present): VALL-E (2023), Bark (2023)

Statistical Parametric Speech Synthesis



Features

- Speech Features

- a. F0 (Fundamental Frequency):

- i. This is basically the pitch of your voice
 - ii. Think of how your voice goes up when asking a question
 - iii. Or how it goes down at the end of a sentence

- b. MCEP (Mel-Cepstral Coefficients), MFCC, Mel-filter bank features:

- i. This will represent the different sound units like vowels, constants, nasals etc
 - ii. Basically it convey the message information

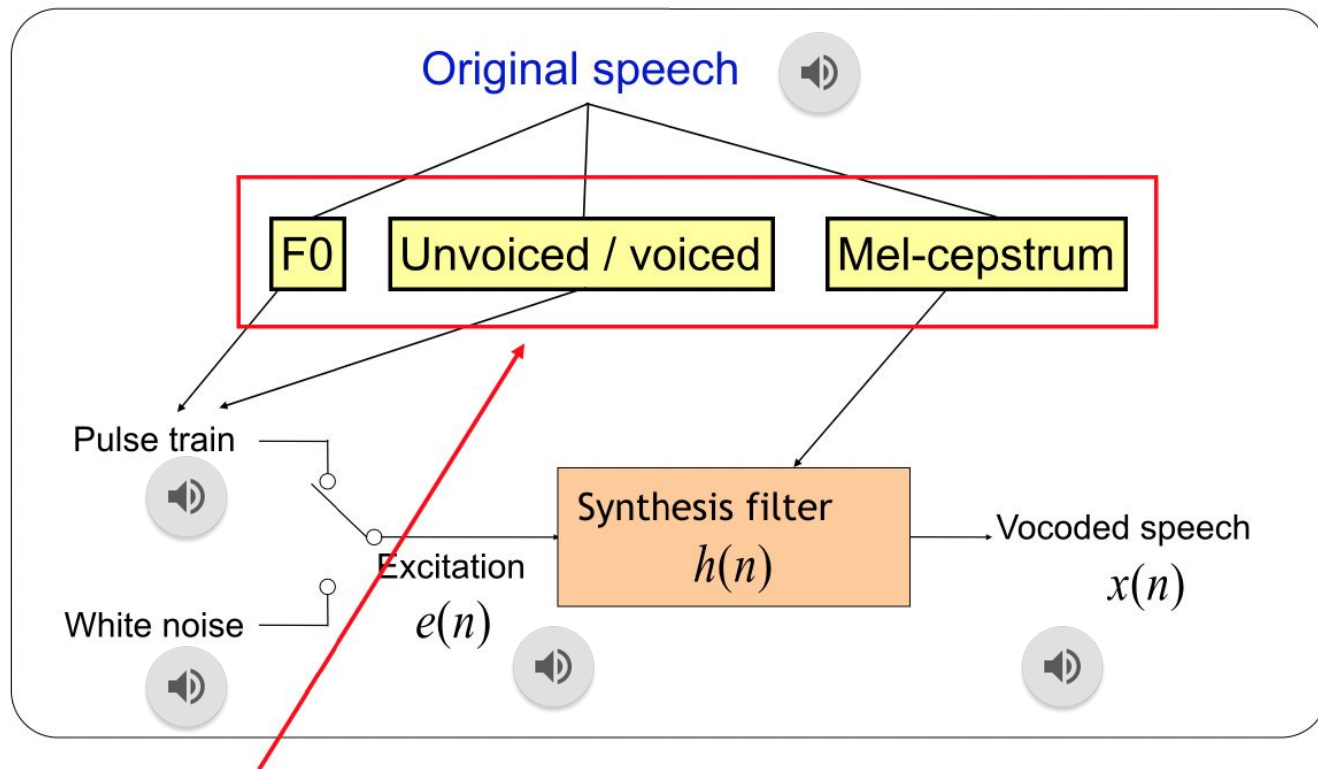
- c. BAP (Band Aperiodicity):

- i. This represents the breathiness or roughness in your voice
 - ii. Like the difference between a whisper and a clear voice

- Linguistic Features

- a. Phone, Syllable, Word, Phrase, Sentence level etc.

Vocoder: Speech Features to Waveform



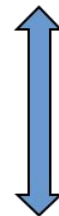
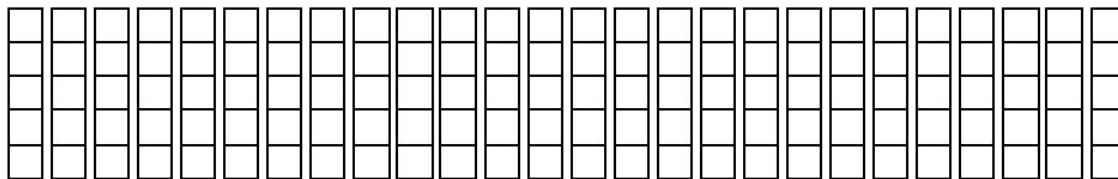
These speech parameters modeled by HMM

Source: HTS web page <http://hts.sp.nitech.ac.jp/>

Sequence-to-sequence regression

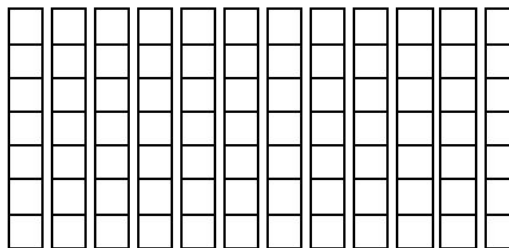
The core problem can be described as sequence-to-sequence regression.

Output sequence:
speech features



Different lengths, because of
different clock rates

Input sequence:
linguistic specification

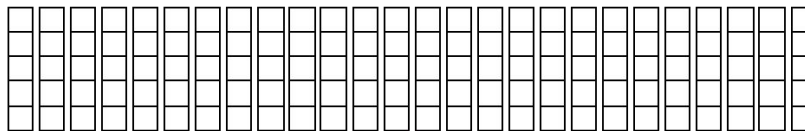


Two tasks to accomplish

Task 1: Sequencing (from phonetic sequence, Decide durations, then Create a sequence of frames)

Task 2: Prediction (Given the local linguistic specification, predict one frame of speech parameters)

Output sequence:
speech features



Regression function

Input sequence:
linguistic specification

```
sil-sil-sil+ao=th@x_x/A:0_0_0/B:x-x-x@x-x&x-x#x-x$...  
sil-sil-ao+th=er@1_2/A:0_0_0/B:1-1-2@1-2&1-7#1-4$...  
sil-ao-th+er=ah@2_1/A:0_0_0/B:1-1-2@1-2&1-7#1-4$...  
ao-th-er+ah=v@1_1/A:1_1_2/B:0-0-1@2-1&2-6#1-4$...  
th-er-ah+v=dh@1_2/A:0_0_1/B:1-0-2@1-1&3-5#1-3$...  
er-ah-v+dh=ax@2_1/A:0_0_1/B:1-0-2@1-1&3-5#1-3$...  
ah-v-dh+ax=d@1_2/A:1_0_2/B:0-0-2@1-1&4-4#2-3$...  
v-dh-ax+d=ey@2_1/A:1_0_2/B:0-0-2@1-1&4-4#2-3$...
```


How the two tasks are solved in HMM based speech synthesis

- The sequencing or alignment task is solved with dynamic programming using Semi-Hidden Markov Models (SHMMs), where the duration of a state is part of the state.
- This task is called force alignment.
- The prediction task is solved by classification and regression that map sets of linguistic labels to Speech features using HMM models.

HMM: Parameter initialization

Input Data: - Speech Waveforms - Corresponding Text Labels

Feature Extraction: -

Extract F0, MCEP, BAP from speech

Extract linguistic features from text

Initial Segmentation: -

Divide speech into states (usually 5 states per phone)

HMM: Baum-Welch Algorithm (EM Algorithm)

E-Step (Expectation): -

Calculate state occupancy probabilities

Determine which frames belong to which states, based on highest likelihood score

M-Step (Maximization): -

Update means and variances

Refine state duration probabilities

HMM: Training example for Vowel “AH”

Initial State: /AH/ = [5,5,5,5,5] frames

Iteration 1: -

Calculate feature likelihoods -

Update duration model -

New assignment: [4,6,7,5,3]

Iteration 2: -

Recalculate with new alignments -

Update duration model -

New assignment: [3,6,8,5,3]

HMM: Final output

Each HMM State has: -

Mean vectors (μ)

Covariance matrices (Σ)

State duration probabilities

State 1 (like 'AH' in "BUS", /b AH s/): -

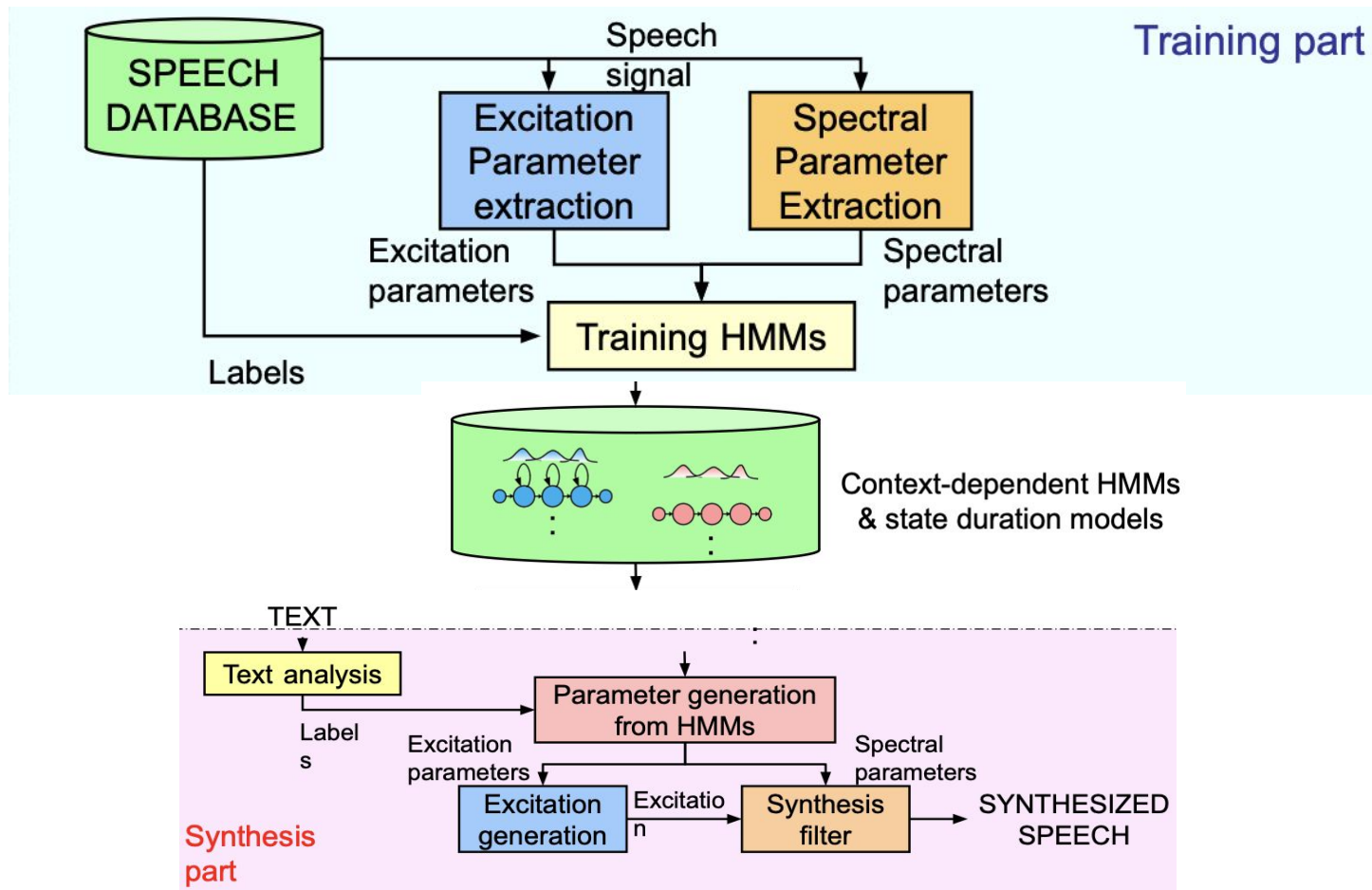
F0 mean: 120Hz

MCEP means: [m1, m2, m3,...]

BAP means: [b1, b2, b3,...] -

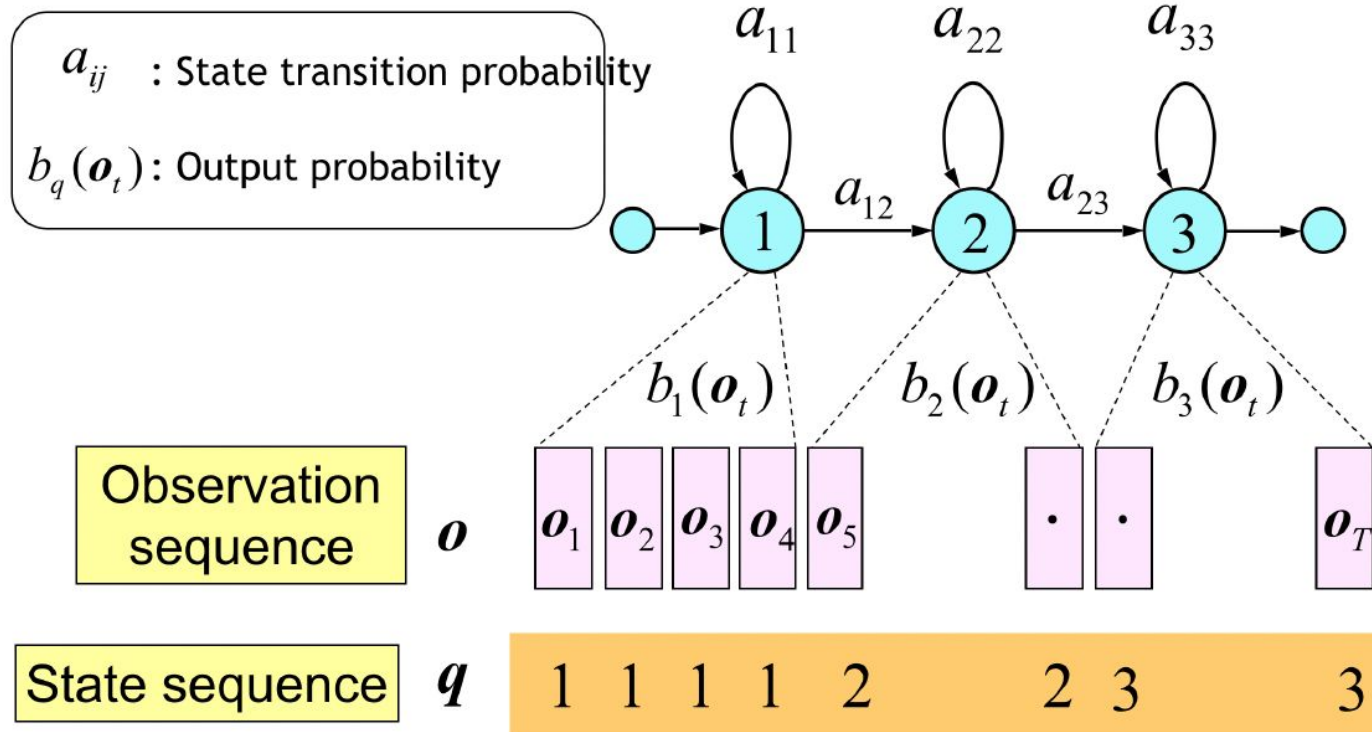
Duration mean: 5 frames

HMM Based Speech Synthesis



Hidden Markov Model

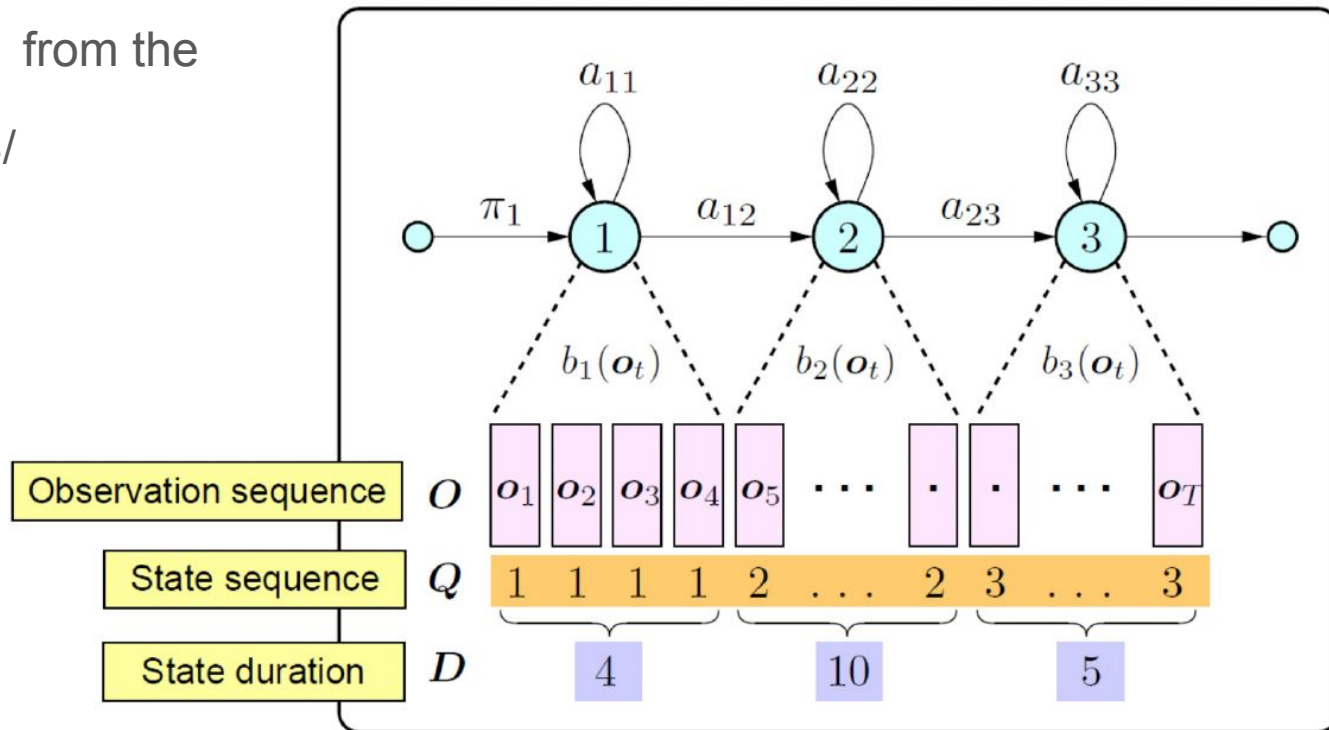
modelling vowel /AH/ from the word /bus/
-> /b AH S/



Source: HTS web page <http://hts.sp.nitech.ac.jp/>

Duration Modeling in HMM

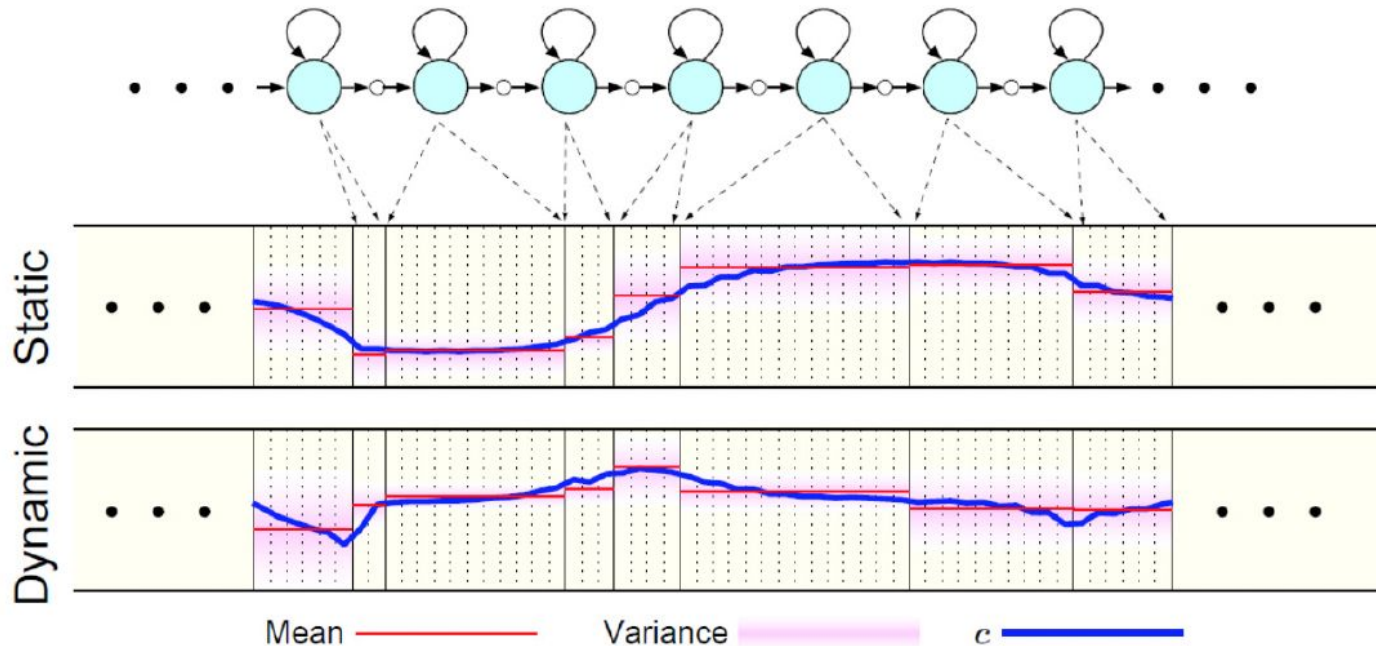
modelling vowel /AH/ from the
word /bus/ -> /b AH S/



Source: HTS web page <http://hts.sp.nitech.ac.jp/>

(*) Tokuda & Zen (2009)

Dynamic parameters



We now have the generated parameters, which we can put through our resynthesis algorithm to convert back to speech!!

Samples: USS



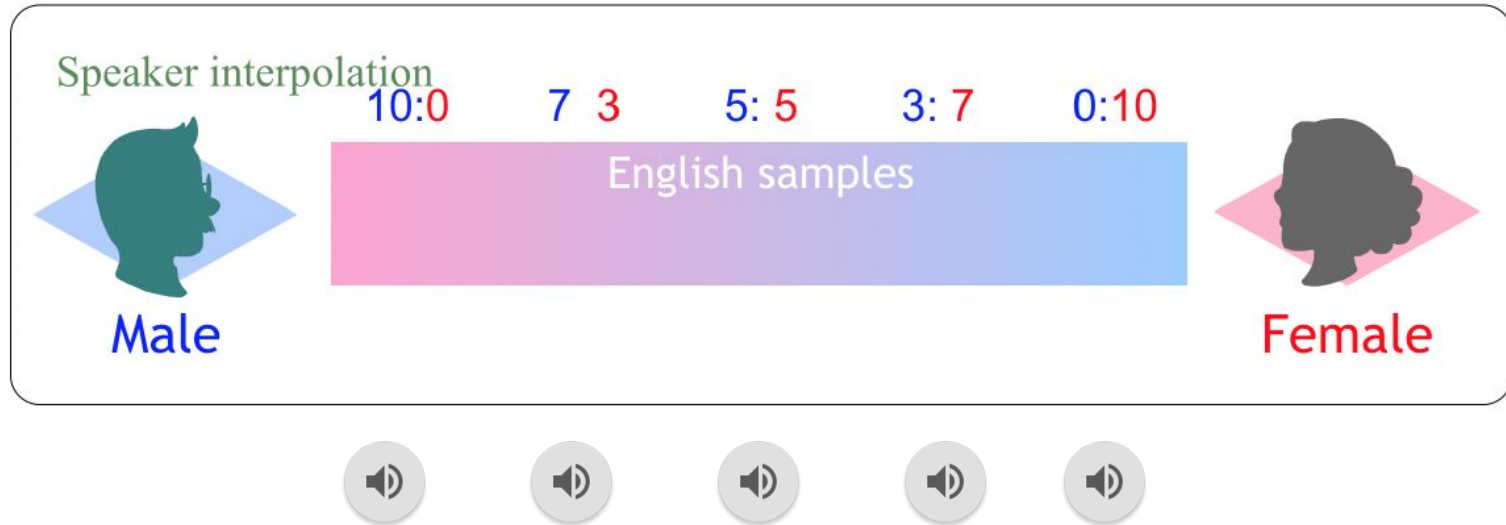
HTS:



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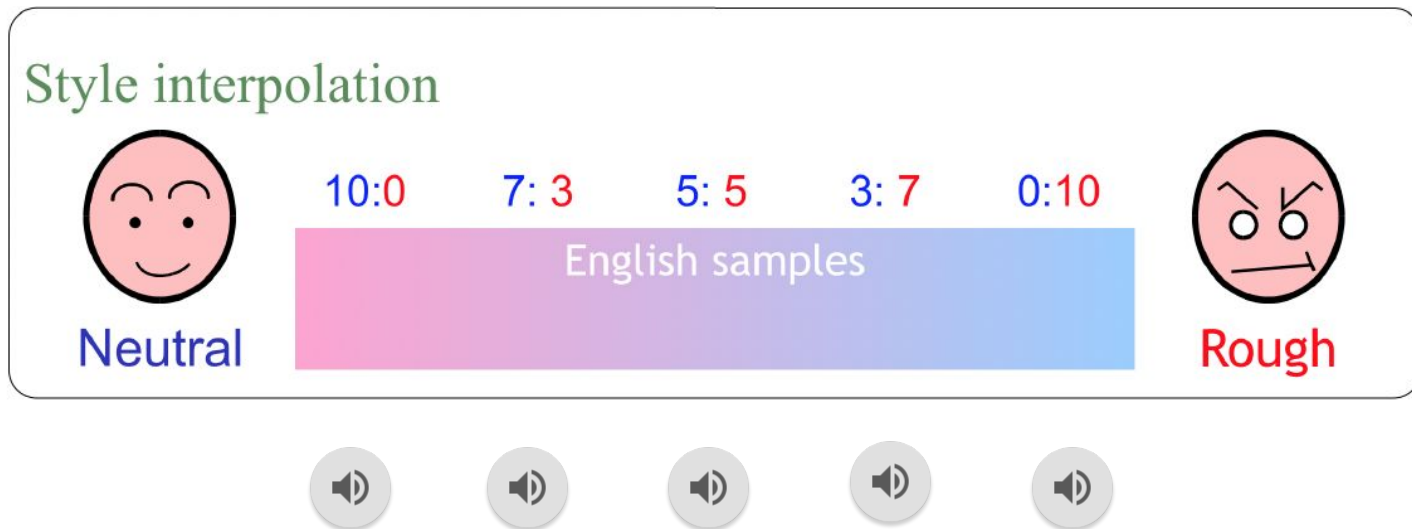
<http://hts.sp.nitech.ac.jp/>

Application 1: Speaker transformation



Source: HTS web page <http://hts.sp.nitech.ac.jp/>

Application 2: Style mapping



Source: HTS web page <http://hts.sp.nitech.ac.jp/>

HMM based Speech Synthesis: Conclusion

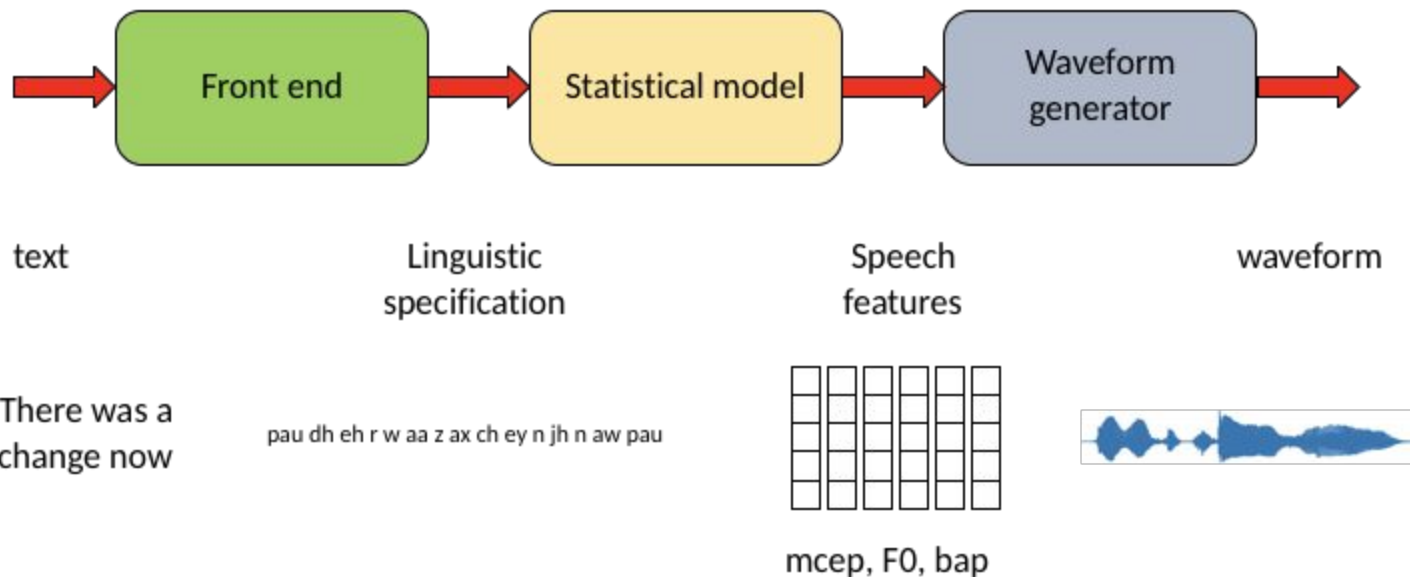
Advantages:

- Small footprint
- Stable output
- Easy to modify voice characteristics

Limitations:

- Over-smoothing of speech parameters
- Buzzy voice quality

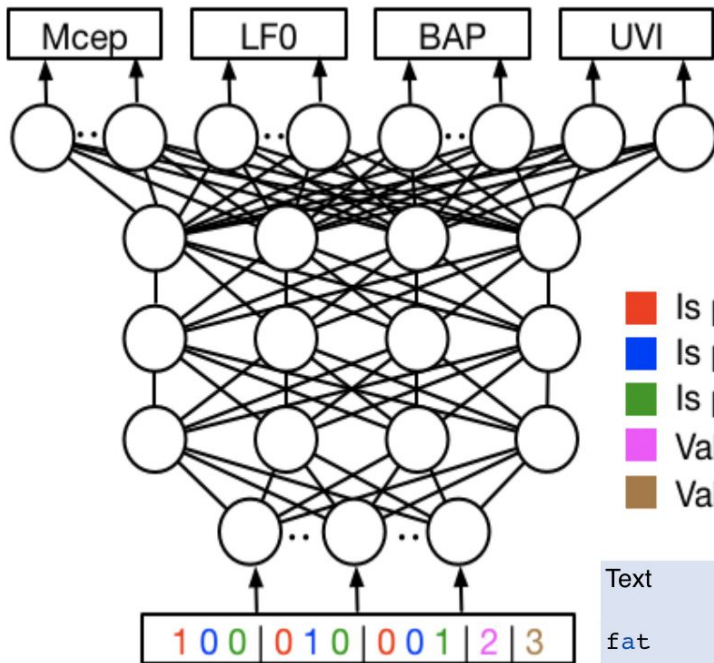
DNN based speech synthesis:



Merlin toolkit link: <https://github.com/CSTR-Edinburgh/merlin>

DNN based speech synthesis:

- Input Layer: Linguist binary/numeric feature
- Output Layer: Acoustic feature



- Is p.ph/c.ph/n.ph **f** ?
- Is p.ph/c.ph/n.ph **ae** ?
- Is p.ph/c.ph/n.ph **t** ?
- Value of ph.pos
- Value of ph.cnt

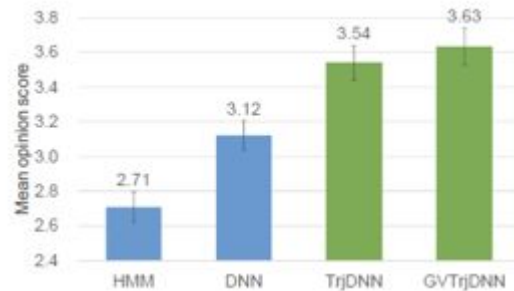
Text	Phone	Label (p.ph+c.ph-n.ph/ ph.pos_ph.cnt)
fat	[f ae t]	L = f+ae-t/2_3

- **Single-DNN**

All streams of the output are combined as a long vector in a single network.

- **Multi-DNN**

Each stream of the output has an individual network.



Merlin toolkit link: <https://github.com/CSTR-Edinburgh/merlin>

References

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- Hunt, A. and Black, A. (1996). Unit selection in a concatenative speech synthesis system using a large speech database, in Proc. ICASSP 96, pp 373-376, Atlanta, Georgia.
- K. Tokuda, Y. Nankaku, T. Toda, H. Zen, J. Yamagishi, and K. Oura, "Speech synthesis based on hidden markov models", Proceedings of the IEEE, vol. 101-5, pp. 12341252, 2013.
- H. Zen, K. Tokuda, and A. W. Black, Statistical parametric speech synthesis, Speech Communication, vol.51(11), pp. 10391064, 2009
- Simon King, An introduction to statistical parametric speech synthesis, Sadhana Vol. 36, Part 5, October 2011, pp. 837{852
- Tokuda, K., Yoshimura, T., Masuko, T., Kobayashi, T., Kitamura, T. Speech parameter generation algorithms for HMM-based speech synthesis" in proceedings of ICASSP 2000.

References

- ISCA Synthesis Interest Group: https://synsig.org/index.php/Main_Page (look there for the SPCC videos)
- CSTR, Simon Kings' page on speech synthesis: <http://www.speech.zone/courses/speech-synthesis/>
- CMU Festvox: <http://festvox.org/>
- HTS web page <http://hts.sp.nitech.ac.jp/> for more information on HMM based speech synthesis system (HTS)
- Google: <https://google.github.io/tacotron/>
- Espnet toolkit link: <https://github.com/espnet/espnet>
- Merlin toolkit link: <https://github.com/CSTR-Edinburgh/merlin>
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Thank You